



OCCAR Management Procedure

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This document replaces: OMP 13_Airworthiness_Issue1_dtd._09/06/15 and OMP 14_Qualification Management_Issue1, dtd. 04/12/2014

Record of changes

Date	Issue	Changes
04/12/14	Issue 1	Initial Issue
18/12/2023	Issue 2	Document reviewed and updated in the frame of a joint review of OMP13 "Airworthiness" and OMP14 "Qualification". Opportunity taken to merge the Airworthiness OMP (OMP13) with the Qualification OMP (OMP14) and to focus on strategic aspects (first and foremost governance) in order to increase the impact of the new merged OMP as a programme enabler. Legacy content and annexes from OMP13 Version 1 and OMP14 Version 1 not kept in Version 2 are captured by OCCAR-EA in Legacy Content Registers on Qualification and Airworthiness. For Version 2, the notion of certification is enlarged to its broader understanding (not branch-related, in particular not only Air-related).

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List of acronyms/definitions/explanations

List of acronyms, definitions and explanations can be found at Annex OMP 14-A.

1. Purpose

The purpose of OMP 14 is to provide governance principles and high-level processes for the management of Qualification and interfaced Certification and Acceptance activities, to enable OCCAR-EA to deliver the programme outputs as specified in the Programme Decisions (ProgD).

The notions of Qualification, Certification and Acceptance are in accordance with their definitions as per Annex A of this document.

OMP 14 addresses the following aspects:

- The fundamental principles to be applied in OCCAR-managed Programmes for the management of Qualification;
- The main steps and contractual principles of the Qualification process, as described in Annex B of this document;
- The roles and responsibilities of the stakeholders in the Qualification process and interfaced Certification and Acceptance activities;
- The interfaces to Certification and related principles to coordinate Certification when applicable;
- The interfaces to Acceptance.

2. Scope

OMP 14 applies to all new OCCAR-managed Programmes¹ where development activities are carried-out and, if beneficial, to new elements of existing OCCAR-managed Programmes. It is to be considered by OCCAR Programme Divisions (PD), or by Programme Integration Teams (PIT).

It provides the principles and procedure for the management of Qualification and interfaced activities (Certification and Acceptance), to be applied to both the Product and its support systems.

For programmes based on Commercial Off-The-Shelf (COTS)/No Development Items (NDI) products, it is assumed that a design fulfilling PS requirements with no or little adaptations needed is already available at the outset of the programme, so that OMP 14 may be applicable only for those sub-projects within the programme where significant design works remain necessary. The ProgD for such COTS/NDI-based programmes shall unambiguously describe in how far OMP 14 remains applicable or not.

3. Applicable Documents

- OCCAR Convention;
- OCCAR Security Agreement;
- OMP 1 Principal Programme Management Procedure;
- OMP 2 Programme Integration;

¹ Nations have access to OCCAR-EA Legacy Repository on Qualification and Certification from OMP13 & OMP14 Issue 1.

- OMP 5 Contract Placement Procedure;
- OMP 6 Contract Terms & Conditions;
- OMP 7 Government Quality Assurance;
- OMP 11.... OCCAR Security Regulations;
- OMP 12.... Handling of Unclassified Sensitive Information

4. Fundamental Qualification Principles

4.1 Qualification principles and main steps

As a general principle, in order to achieve best value-for-money for the Qualification of Products acquired via OCCAR-managed Programmes, Participating States (PSs) should systematically seek to use international standards², common procedures and cooperative work as widely as possible. In particular, the aim is to adopt recognition whenever appropriate in order to allow the re-use to the largest possible extent the work achieved by other parties. In addition, Participating States should largely delegate to and empower the OCCAR Programme Division and (as applicable) a Qualification management structure.

Qualification defined in this OMP should be tailored to meet the individual OCCAR-managed Programme requirements. Nevertheless, programmes shall adhere to the governance and contractual principles laid down at Annex B to this document.

4.2 Principles of communality, synergy and recognition

While establishing the detailed and tailored Qualification process to be applied by an OCCAR-managed Programme, OCCAR will endeavour to maintain or even increase the effectiveness and efficiency of the Qualification process. For this purpose, harmonisation of Qualification, Certification and Individual Acceptance (QCA) Plans is required early in the programme with the objectives of avoiding unnecessary duplication and seeking cost-effectiveness.

PSs will apply international standards and procedures whenever possible (e.g. ITOPS/STANAGs/EMARs etc).

Wherever possible and appropriate, PSs should agree on harmonized and common processes and aim at recognizing other national organizations (National Qualification and/or Certification Authorities) in order to re-use artefacts produced by their respective national processes.

4.3 Efficient and empowered Qualification management structure

OCCAR-EA is responsible to manage the Qualification process within a programme. However a dedicated organisation may be created to assist Qualification and/or Certification and to ensure that the expertise available within the PSs is effectively used. In this document hereafter, such organisation is referred to as Certification and Qualification Organisation (CQO). All principles laid down for the CQO remain applicable for similar organisations named otherwise.

² PS should in particular consider European standards (<https://edstar.eda.europa.eu>)

A CQO, mutually accepted between OCCAR, PSs and potentially with the Contractor, should be specifically established for the programme wherever possible in order to optimise a common application of Qualification requirements. If established, the requirement for a CQO shall be defined in the ProgD. The ProgD shall provide a clear allocation of responsibilities between OCCAR-EA and the authorities (including their support to the CQO) and the decision-making process. Generally, the CQO will manage Certification and Qualification according to Terms of Reference endorsed at Programme Committee (PC) level. OCCAR shall remain responsible to ensure that all contractual outputs needed for the purpose of Certification are delivered according to the needs of the programme (in particular for the CQO if applicable and Certification Authorities).

For new aeronautical programmes, owing to specific airworthiness risks, the establishment of a CQO is most strongly recommended and any exception to this principle shall be formally endorsed by the Programme Board and as needed by the Board of Supervisors.

Special attention is required when a CQO is set up for a programme. In accordance with the respective ProgD, OCCAR-EA shall be responsible for managing the Qualification process as defined in the programme Contract. The ProgD shall clearly indicate which option from the two below is applicable, as well as potential deviations:

- **Option 1: OCCAR-EA is fully responsible for the management of the Qualification Process**

Under this option, the Programme Manager (PM) will ensure the PSs have transparency access to the documentation exchanged during the Qualification Process and to any findings resulting from qualification related to performance, schedule and cost affecting them.

The PSs will, upon request from the PM, provide advice from subject matter experts and assist the qualification process. Any requests from the PM for assistance from subject matter experts will be addressed to the National Programme Coordinators (NPCs). The contributions of the PSs to the Qualification Process should be coordinated within the OCCAR Programme Working Group (PWG) unless the PC decides otherwise. Upon completion of the Qualification, the PM submits his recommendation related to the acceptance of the Qualification to the PC for approval. For Certification, the PM will ensure to assist and provide all necessary support to the certification stakeholders (in particular Certification Authorities).

- **Option 2: CQO provides advice to OCCAR-EA which remains fully responsible for the management of Qualification process**

Under this option, a CQO is created in order to facilitate the qualification and certification process. The PSs should commit to support the CQO with the required resources in terms of qualified experts, funding and facilities. The CQO manages Certification and Qualification activities in close coordination with OCCAR-EA and will provide its advice and conclusions to the PM and to the relevant certification bodies as applicable.

The PM takes the decisions related to Qualification based on the CQC's advice, aiming at the CQC's recommendation of Qualification and Certification achievement.

5. Certification activities

Certification is a national prerogative. However, being part of an international cooperation, it is essential that PSs define a set-up based on economy of effort and cooperation. To do so, PSs shall commit to the following principles in the ProgD:

Each Nation is responsible for the regulation of its own Products. As a result, all Certification activities within OCCAR PSs are conducted and regulated on a national basis. To minimise time, cost and any associated delay risks, it is essential that PSs reach agreement on the fundamental certification principles.

For OCCAR programmes these Certification principles are:

- The certification management processes shall be defined by OCCAR-EA, the Certification Authorities, the CQO (if formed) and the Contractor prior to agreement between OCCAR-EA and the Contractor. The processes may be tailored to suit the needs of the programme.
- The processes of certification and of qualification are based mainly on the same body of evidence: studies, simulations and tests done in operational environment. Therefore, to increase the efficiency of the programme, certification and qualification shall be integrated as much as possible to find synergies between the programme development, certification and qualification activities. These activities shall be performed with an intention to minimise effort and avoid duplication of resources and activities. To save time and money, Certification authorities should agree on common and harmonized procedures and way of working, and should consider recognition of the other Participating States' Certification Authorities, with the exception of those areas whereby the re-use of artefacts to avoid duplication of effort cannot be reasonably ensured.
- Although PSs remain responsible for the Certification of their Products, OCCAR-EA shall support and coordinate the Certification process, ensuring in particular that the Contract enables to effectively produce the body of evidence needed by the Certification Authorities to grant Certification of the Product. An open and collaborative communication with OCCAR is mandatory to enable to perform an efficient management of contractual deviations related to Certification
- All PSs shall rely on international military and civil certification standards and organisations for certification activities to the maximum extent possible. Where a civil certification organisation e.g. European Union Aviation Safety Agency (EASA) is required to contribute to the overall certification process, the CQO, the civil certification organisation and the contractor should discuss and agree the process in order to establish an effective cooperation to the greatest extent for verifying that the military certification requirements are met.
- All PSs shall commit to exchange all relevant information (including to the largest possible extent that of a classified or sensitive nature and in accordance with OMP 11 and OMP 12) in order to allow the PSs to fulfil their national certification obligations;

- All PSs shall provide personnel who are competent³ to undertake their assigned tasks. If a CQO is established, it will provide recommendation to the Certification Authorities concerning the issue of the Certification statement⁴ or equivalent and the approval of any modifications under their respective processes. The Certification Authorities shall therefore uphold and accept the output of the certification-related activities performed by the CQO without the routine need for further investigation. The CQC involvement does not modify the rights and legal responsibilities of the Certification Authorities or affect national certification regulations;
- Certification management structures (CQO or equivalent international or national body) shall strive to maintain an open and collaborative communication with OCCAR, in order to enable this latter to perform an efficient management of contractual deviations related to Certification.
- For post-certification activities, specific programme arrangements may be developed, governed by the goal to manage post-certification activities as much as possible under common and harmonized procedures and way of working between Nations and in particular Certification Authorities. PDs will provide their active support and assistance to facilitate such cooperative post-certification arrangements.

PSs shall commit to the above principles in the ProgD.

6. Acceptance activities

As a general principle, the availability of a Qualified Design (see Annex B, Step 3) is a pre-requisite to conduct Individual Acceptances⁵, unless otherwise settled with the PSs.

The purpose of the Individual Acceptance step is to verify that each individual series production item meets the contractual requirements using the agreed acceptance criteria. The individual item acceptance will be based on a Certificate of Conformity (CoC) or equivalent document and the assessment of the Acceptance tests and, if Certification is a pre-requisite to delivery, on a formal certification document as detailed in the contract. The acceptance tests should be tailored to each specific Product type and define the relevant level of testing (e.g. individual test for aircraft, tests on samples only for munitions, submarine campaigns, sea trials, ...).

Governmental Quality Assurance (GQA) activities as required within the Product Acceptance process will be performed by the GQAR in accordance with OMP 7.

7. Principle Roles and Responsibilities

Decisions related to Qualification, Certification and Acceptance are taken in accordance with the roles and responsibilities within an OCCAR-managed Programme as defined in the OCCAR Management Procedures and the ProgD(s).

7.1 The Programme Board

The Programme Board:

³ According to national available human resources, PSs shall provide suitably qualified, experienced and empowered technical personnel.

⁴ e.g. Military Type Certificate (MTC) for Aeronautical Programmes

⁵ For aeronautical product this could include release certificates issued by approved production organizations to state the airworthiness of products when required

- Ensures that Qualification and Certification principles laid down in this OMP (in particular: collaborative approach and empowered Qualification and Certification management structure) are reflected in the ProgD;
- Ensures the early engagement of the National Military Certification Authorities prior to signature of the ProgD;
- Ensures that the fundamental Certification principles are reflected in the relevant ProgD(s);
- Actively supports and sponsors national processes leading to timely national Certification and ensures that the most cooperative and efficient way of certifying has been reached, fostering harmonized and common processes and re-use of available artefacts whenever appropriate.

7.2 The Programme Committee

The PC will:

- Endorse the Qualification Programme Plan (QPP), according the schedule and provision of required national resources, prior to approval by the PM;
- Act on behalf of the Programme Board to ensure that the fundamental Certification principles (see § 7.1 above) are applied;
- When a CQO or equivalent structure is established, approve its organisational structure and Terms of Reference (ToR), appoints the CQC Chairperson⁶ based on a recommendation from the Certification Authorities and from the National Military Qualification Authorities (NMQAs) and oversees the CQO management;
- Endorse Qualification and/or Certification documents (issued by the CQO if a CQO is established) prior to approval by the PM;

7.3 OCCAR-EA Director

The OCCAR-EA Director is responsible for achieving the Programme High Level Objectives (HLOs). They delegate the management of Qualification to the PM.

7.4 The Programme Manager

The PM is the Head of the Programme Division and mandated by the OCCAR-EA Director to manage the programme. The PM will approve the documents delivered in the Qualification steps and provide the PSs with the required documents related to Qualification.

7.5 The Certification and Qualification Organisation

The CQO may include OCCAR members and representatives from national authorities as well as other representatives if relevant.

The PSs under the ProgD should commit to provide the required resources in terms of experts, information and funding to the CQO. The CQO evaluates the results of the

⁶ The CQC Chairperson will be appointed by the Certification Authorities of the PSs where no Programme Board is established.

Qualification process making sure that the national authorities and experts are involved as required. The CQO comprises of the Certification and Qualification Committee (CQC) with a chairperson and a number of expert panels composed of subject matter experts from national authorities. The chairperson of the CQC may be an independent appointee by the PSs, the decision to have an independent chairperson should be based upon the complexity of the programme and the technical and financial benefits but rests with the PSs. The CQO is responsible for validating the Certificate of Design (CoD). The CQC chairperson provides a report on the Qualification to the Programme Manager (PM) who will issue a qualification statement for approval to the Programme Committee (PC).

All PSs shall provide personnel who are competent⁷ to undertake their assigned tasks and shall therefore recognise, uphold and accept the output of the certification⁸ and qualification activities performed by the Certification & Qualification Organisation (CQO) without the routine need for further investigation.

7.6 Contractor

The Contractor is responsible for achieving the programme requirements (in particular the Qualification and Certification of the Product or system) as defined in the Contract through contractual requirements.

7.7 Certification Authorities

The Certification Authorities of the PSs shall:

- Propose and advise the Programme Board regarding Certification HLOs to be included as HLOs in the ProgD.
- Based on recommendation of the CQC when applicable, issue the formal authorizations and certificates necessary according to the applicable regulation during the development, production and in-service support of the Product(s), both for their own national Product(s) and for common cooperative activities under the ProgD;
- Commit to providing all information as requested by law in the chosen certification framework, including accident, incident and occurrence data to the other Certification Authorities of PSs and Industry in order to fulfil their national Certification obligations;
- Contribute to the Certification management activities allowed under the terms and conditions of the Contract throughout the whole life of the programme;
- Contribute to the overall identification of Certification risks, assessments and mitigation plans;
- When a CQO is established:
 - Endorse the organisational structure and the ToR for the CQO;

⁷ According to national available human resources, PSs shall provide suitably qualified, experienced and empowered technical personnel.

⁸ The CQC involvement does not modify the rights and legal responsibilities of the Certification Authorities or affect national certification regulations

- Provide competent⁹ personnel to the CQO;
- Recognise, uphold and accept, without the routine need for further investigation, all certification-related activities made by, or information, data etc provided by the CQO;
- Delegate to the CQC to the maximum extent possible the following activities (unless delegation is forbidden by national regulation):
 - Ensure that Certification requirements for the programme meet the essential requirements for Certification applicable to the Product;
 - Establish the Certification requirements to be included in the Contract and confirm the relevance and completeness of those requirements to OCCAR in consultation with (and whenever applicable, with the approval of) Certification Authorities;
 - Define and propose to the Certification Authorities the Certification High Level Objectives to be included in the ProgD.
 - Define the overall structure of the CQO (including the number, type and composition of the CQ Panels) and establish ToRs;
 - Establish and approve all the Certification Requirements to be included in the Contract.
 - For Aeronautical programmes in particular¹⁰:
 - Approve the Certification Base (CB)¹¹ for inclusion in the Contract¹²;
 - Contribute to the assessment of the flight conditions, supporting the Military Permit to Fly
 - Harmonize amongst Certification Authorities the organisational requirements for design, production and maintenance organisations¹³ (for Industry) throughout the whole life of the programme and ensure that they are reflected within the relevant contract(s);
 - Confirm to OCCAR as far as Airworthiness is concerned the Programme Information Document (PID) describing the organisational structure, CQ processes and procedures to be applied within the respective programme in order to support the successful execution of the programme;

⁹ According to national available human resources, PSs shall provide suitably qualified, experienced and empowered technical personnel.

¹⁰ (in such case, the Certification Authorities are designated as National Military Airworthiness Authorities, NMAAs)

¹¹ It should be noted that the initial (Type) Certification Basis may need to be changed along the course of the certification process due to new applied technologies, introduction of design changes, discovery of unsafe conditions or compliance demonstration results.

¹² NMA approval maybe required before initial CB finalisation.

¹³ Based on recommendation from CQC the certification authorities approve Organizations and grant associated privileges when relevant.

- Approve appropriate templates (e.g. EMAR forms) as required.

7.8 Other Stakeholders

Depending on the programme and national regulations in the PSs, as well as the programme type, specific stakeholders may be involved in the qualification process. These are (non exhaustive list):

- National Design Approval Authority (NDAA);
- Government Quality Assurance Authority (GQAA);
- National Test Centres;
- Safety Authorities;
- Security Authorities;
- Accreditation Authorities;
- Ammunition Authorities;
- Bodies issuing standards applicable to the qualification, in particular NATO standardisation groups.

8. **Annexes**

Annex OMP 14-A	Acronyms, Definitions and Explanations
Annex OMP 14-B	Qualification main steps and contractual principles